

The protective effects of meaning in life on the relationships between attachment anxiety, rumination, and nonsuicidal self-injury: A moderated mediation model

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ABSTRACT

Nonsuicidal self-injury (NSSI), which refers to the direct and deliberate destruction of one's body tissues without the intention of death, is a serious public health problem. Attachment anxiety is an important predictor of NSSI, but the underlying mechanisms between their relationship need to be further explored. This study examined the mediating role of rumination and the moderating role of meaning in life in the relationship between attachment anxiety and NSSI.

Methods: The study was cross-sectional, and 2,716 Chinese junior high school students (53.3 % male; $M_{age} = 12.25$ years, $SD = 0.86$) were measured by the attachment anxiety subscale of Experience of Close Relationships-Relationship Structures Scale, the Rumination Responses Scale, the Self-Injurious Behavior Assessment Questionnaire, and the Meaning in Life Questionnaire.

Results: The results showed: (1) attachment anxiety, rumination and NSSI were significantly and positively correlated; (2) rumination partially mediated the relationship between attachment anxiety and NSSI; (3) meaning in life moderated the pathways of attachment anxiety $\rightarrow$ NSSI and rumination $\rightarrow$ NSSI respectively.

Conclusions: Findings of the study further reveal the mechanism underlying the relationship between attachment anxiety and NSSI from a cognitive perspective, and also help provide ideas for the prevention and intervention of adolescents NSSI (i.e., improving adolescents' meaning in life).

1. Introduction

Nonsuicidal self-injury (NSSI) refers to the direct and deliberate destruction of one's body tissues without the intention of death (Nock, 2009). Common forms of NSSI include self-cutting, burning, carving to bleeding, scratching skin to bleeding, etc. (Gong et al., 2019). NSSI usually occurs and peaks during adolescence (Nock, 2010). The lifetime NSSI prevalence was about 17.2 % in adolescents over the world (Swannell et al., 2014), and it is still on the rise (Paul et al., 2015;

Wester, Trepal, & King, 2018). In Chinese adolescents, the 12-month prevalence of NSSI was 26.4 % (Gong et al., 2019) or even as high as 41.5 % (Wang et al., 2016). Moreover, adolescent NSSI has become a serious public health problem because of its significant associations with psychological disorders (Zhu et al., 2021) and suicidal attempts (Hamza, Stewart, & Willoughby, 2012; You & Lin, 2015). Given the high prevalence and clinical significance of NSSI, it is important to clarify its developmental mechanisms for effective prevention and intervention.

Attachment theory is one of the most influential conceptual

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frameworks for interpreting the development of maladaptive behaviors (Al-Yagon, 2015; Caspers et al., 2005). In this study, we are also going to examine the link between attachment and NSSI. Researchers have identified two dimensions of attachment: attachment anxiety and attachment avoidance (Bartholomew & Horowitz, 1991; Hazan & Shaver, 1987). Previous studies have suggested that higher attachment anxiety, instead of attachment avoidance, is associated with higher risks of NSSI (Woo, Wrath, & Adams, 2020; Tatnell et al., 2014). Therefore, we will focus on the relationship between attachment anxiety and NSSI.

1.1. The relationship of attachment anxiety to NSSI

Attachment anxiety refers to individuals' fear of being rejected and abandoned by their attachment figures (Hazan & Shaver, 1987). Individuals with attachment anxiety desire to keep intimacy with their attachment figures (Joel, MacDonald, & Plaks, 2012; Lin et al., 2020). Besides, they have heightened sensitivity to potential signals indicating threats to their attachment relationships (Ávila et al., 2015), and may thus experience elevated negative emotions (Burnette et al., 2009). These characteristics make them prone to engage in NSSI as a method to maintain relationships or regulate emotions. In fact, previous research have supported the relationship between attachment anxiety and NSSI (Levesque et al., 2021; Woo, Wrath, & Adams, 2020; Molaie et al., 2019).

Although attachment anxiety could contribute to NSSI, the psychological mechanisms underlying the relationship are largely unclear (Wrath & Adams, 2019). The Integrated Motivational-Volitional (IMV) model, which was initially proposed to explain the emergence of suicide (O'Connor, 2011), has also been applied to understand the development of NSSI (O'Connor, Rasmussen, & Hawton, 2012). According to the IMV model, early individual difference characteristics (e.g., attachment anxiety) may make individuals vulnerable to proximal risk factors (e.g., rumination) and then trigger NSSI (Tatnell et al., 2014), and motivational factors (e.g., meaning in life) may play a protective role in the process. Therefore, we would like to explore the mediating role of rumination and the moderating role of meaning in life in the relationship between attachment anxiety and NSSI.

1.2. Rumination as a mediator

Rumination is defined as the tendency to excessively rethink past negative experiences (Nolen-Hoeksema, Wisco, & Lyubomirsky, 2008). As a negative cognitive style (Liu et al., 2021), rumination may keep individuals entrapped in negative experiences (Nolen-Hoeksema, 1991) and hinder them from positively coping (Turan, Hoyt, & Erdur-Baker, 2016). Thus, rumination may contribute to some negative psychological outcomes, such as depression and aggressive behaviors (Sun et al., 2014; Pedersen et al., 2011).

According to the IMV model, the high sensitivity to negative signals would render individuals entrapped in negative cognitions (e.g., rumination, O'Connor & Kirtley, 2018). Because of the fear of being abandoned, individuals with attachment anxiety may have heightened sensitivity to negative signals and pay more attention to the potential threats in the environment (Griffin & Bartholomew, 1994), leading them to ruminate over these negative experiences (Rognoni et al., 2008). Empirical studies supported the relationship between attachment anxiety and rumination (Garrison et al., 2014; Saffrey & Ehrenberg, 2007). For example, a recent study found that university students with attachment anxiety were more likely to engage in rumination (Bugay-Sökmez et al., 2021).

Meanwhile, rumination could contribute to NSSI. In the IMV model, rumination is associated with "entrapment", which is the key premise of increasing NSSI risk (O'Connor et al., 2012). Rumination causes individuals to focus on the negative aspects of life, leading them to be entrapped in the vicious loop of distress (O'Connor & Kirtley, 2018). To distract themselves from the vicious loop, individuals may choose to

engage in NSSI, because the physical pain caused by NSSI could instantly distract them from negative emotions and cognitions (Gu et al., 2020). Recent empirical studies have also supported that rumination may contribute to NSSI. For instance, Ying et al. (2021) and Buelens et al. (2019) both found that there was a significant positive association between rumination and NSSI.

Extensive studies have supported the notion that attachment anxiety is related to rumination, and rumination is related to NSSI. To our knowledge, however, no previous studies have directly confirmed the mediating role of rumination in the relationship between attachment anxiety and NSSI.

1.3. Meaning in life as a moderator

The majority of research have explored the risk factors for NSSI (Jarvi et al., 2013; Fox et al., 2015); few studies have explored the potential protective factors. We are going to examine meaning in life as a protective factor in this study. Meaning in life refers to one's own subjective purposes or goals (Glaw et al., 2017). In the IMV model of NSSI (O'Connor & Kirtley, 2018), researchers proposed that active goal pursuit (a typical characteristic of meaning in life; Newman & Nezelek, 2019; Park, 2010) could buffer against the emergence of NSSI. Those who believe their life is meaningful tend to broaden their attentional scope (Kok & Fredrickson, 2013) and see alternatives to cope with the current distress, thus adopting adaptive strategies to overcome the current distress (Miao, Zheng, & Gan, 2017). Thus, having high meaning in life, individuals with attachment anxiety or individuals with rumination may use more positive ways instead of NSSI to alleviate their negative emotions or thoughts.

Meaning in life has been demonstrated its protective role in decreasing several adolescent maladaptive behaviors (e.g., internet addiction, suicide; Zhang et al., 2015; Kleiman et al., 2013), but its protective effects on NSSI remain unclear. Therefore, in the current study, we would like to examine the protective role of meaning in life in the relationship between attachment anxiety and adolescent NSSI, as well as in the relationship between rumination and adolescent NSSI.

1.4. The current study

Although empirical studies have provided evidence for the association between attachment anxiety and NSSI, the current study aims to make progress in the following aspects. First, the mediating mechanisms underlying the relationship between attachment anxiety and NSSI remains largely unclear. Previous studies have mainly incorporated emotional factors (e.g., emotional dysregulation, emotional pain; Guérin-Marion, et al., 2021; Molaie et al., 2019) as mediators. However, according to the IMV model of NSSI (O'Connor & Kirtley, 2018), being entrapped in negative cognitions (i.e., rumination) is important in the development of NSSI. Thus, it is necessary to explore the potential mediating role of rumination in the relationship between attachment anxiety and NSSI. Second, previous studies have mainly explored the mediating mechanisms between attachment anxiety and NSSI, while the exploration of protective factors is relatively limited. Therefore, while proposing the mediation model, we also aim to verify the protective effect of meaning in life. Third, considering that NSSI usually occur in early adolescents (11–14 years; Adrian et al., 2019; Wang et al., 2017), the current study would focus on secondary school students to investigate the mediating role of rumination and the moderating role of meaning in life.

In short, we are going to examine a moderated mediation model among Chinese secondary school students to better explain the relationship among attachment anxiety, rumination, meaning in life, and NSSI. Specifically, on the basis of the literature review, we hypothesize that: (1) attachment anxiety and NSSI are positively correlated; (2) rumination will mediate the relationship between attachment anxiety and NSSI; and (3) meaning in life will moderate the mediation model.

2. Method

2.1. Participants

Participants were recruited from two secondary schools in Hunan Province, China. A total of 2,726 adolescents were originally collected. But due to that 10 students had more than 50 % missing values in their questionnaires, only 2,716 (53.3 % male; $M_{\text{age}} = 12.25$ years, $SD = 0.86$) were retained in the study. Among all the participants, 86.0 % came from intact families, 7.9 % came from single-parent families, 4.7 % came from reconstituted families, and 1.5 % came from other family structures (e.g., living with their grandparents). Approximately 24.7 % of the students are the only child, and the rest have siblings. In terms of family economic status, 70.2 % of the students believed that their family economic status was at an intermediate level relative to their classmates, and 1.6 % of the students believed that their family was the poorest among their classmates. When it came to the education level of parents, 52.7 % of the students' fathers and 63.5 % of the students' mothers had only junior high school education or below.

2.2. Procedure

Throughout the investigation process, we obtained the oral consent of the principal and the teachers, as well as the written consent of the students and parents. Before collecting the data, the researchers trained the teachers as experimenters to ensure that they can follow the standard procedures in the collecting process, such as how to conduct a structured guidance phrases and handle emergencies. During the assessment process, students were informed of the anonymity of their filling process and the confidentiality of their answers, as well as the right to stop filling in at any time and for any reason. Especially, we guaranteed to keep students' fill-in results confidential to the school and families and only used the data for research. The students completed a battery of online questionnaires in their school computer classes under the supervision of trained teachers.

All study materials and procedures were approved by the ethics committee of the first author's university.

2.3. Materials

2.3.1. Attachment anxiety

Attachment anxiety between adolescents and their parents in this study was measured by the attachment anxiety subscale of the Experience of Close Relationships-Relationship Structures Scale (ECR-RS, Fraley et al., 2011). The subscale has three items (i.e., "I am afraid that my parents will abandon me", "I am worried that my parents will not care about me as much as I care about them", and "I often worry that my parents do not really care about me"), which were rated on a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree). The ECR-RS has shown good reliability and convergent validity in previous studies (Karataş & Demir, 2019; Donbaek & Elklit, 2014), as well as its applicability to Chinese adolescents (Zhang et al., 2022; Flykt et al., 2021). The Cronbach α of this subscale was 0.81 in this study.

2.3.2. Rumination

This study used the Rumination Responses Scale (RRS; Treynor, Gonzalez, & Nolen-Hoeksema, 2003) to measure rumination. This scale includes two dimensions (i.e., brooding and reflection), and each dimension has five items rated on a 4-point scale, ranging from 1 (*never*), 2 (*sometimes*), 3 (*often*), to 4 (*always*). The scale has been proved sufficient concurrent and overtime validity (Gu, Ma, & Xia, 2020; Gong et al., 2019), as well as its applicability to Chinese adolescents (Zhang et al., 2021; Liang & Lee, 2019). The Cronbach's α of this scale in this study was 0.86.

2.3.3. Nonsuicidal self-injury (NSSI)

In this study, we measured 12 NSSI behaviors (including self-cutting, burning, carving word or pictures on the skin, scratching the skin, inserting objects to the nail or skin, hair pulling, biting, erasing skin, dripping acid onto skin, scrubbing skin using bleach or cleaner, hitting the head or other body parts violently to cause bruising, and punching). The question presented in the questionnaire is "In the past 6 months, have you engaged in the following behaviors to deliberately harm yourself, but without suicidal intent?" The above 12 behaviors were rated on a 6-point scale, ranging from 0 (*never*), 1 (*once*), 2 (*twice*), 3 (*three times*), 4 (*four times*), to 5 (*five times or more*). The reliability and validity of the scale for Chinese adolescents were good (Gong et al., 2019; Ren et al., 2019). The Cronbach's α of this scale was 0.93 in this study.

2.3.4. Meaning in life

The current study used The Meaning in Life Questionnaire (MLQ; Steger et al., 2006) to assess the individual's meaning in life. The scale includes two dimensions (i.e., presence of meaning and search for meaning), and each dimension has five items. The items of the presence of meaning in life (POM) subscale are like "I know the meaning of my life well", and the items of the search for meaning (SFM) subscale are like "I am looking for something that makes my life meaningful". All items were rated on a 7-point Likert scale, ranging from 1 (*strongly disagree*) to 7 (*strongly agree*). The MLQ has shown great reliability and validity (Yek et al., 2017; Ye et al., 2019) in Chinese adolescents (Chen et al., 2021; Chen & Gao, 2021). The Cronbach's α of the total scale in this study was 0.75.

2.4. Analysis plan

Before analysis, the distribution of each variable was checked for normality. Only the skewness and kurtosis levels of NSSI were not within the acceptable ranges (i.e., skewness < 3 and kurtosis < 10), so the log transformation (skewness = 1.97, kurtosis = 3.03) was used to adjust them (Kline, 2016). The transformed variable of NSSI was used in the following analysis.

First, the descriptive statistics and correlation analysis among all variables were calculated through SPSS 22. Second, we verified the mediating role of rumination between attachment anxiety and NSSI through the model 4 in the PROCESS macro by Hayes (2013). A bias-corrected bootstrap method based on 5000 samples was used to estimate the standard error of the indirect effects. The 95 % bias correction (BC) confidence interval (CI) excluding zero indicated the significance of the pathways. Third, we conducted a moderated mediation analysis to investigate whether the mediating path can be buffered by meaning in life through the model 15 in the PROCESS macro. When the interaction effect was significant, simple slope analyses would be conducted.

Before all analyses, all variables were standardized.

3. Results

3.1. Descriptive statistics

The 6-month prevalence of NSSI was 29.7 % ($n = 808$) in our sample of Chinese adolescents. Among all self-injurers, 37.7 % ($n = 305$) reported using only one method, and 62.3 % ($n = 503$) reported using multiple methods. In addition, only 19.8 % self-injurers ($n = 160$) engaged in NSSI only once, while 80.2 % ($n = 273$) engaged in NSSI twice or more. Among the 12 methods of NSSI, carving word or pictures on the skin (65.5 %, $n = 529$) was the most prevalent method, followed by self-cutting (38.4 %, $n = 310$), hair pulling (36.1 %, $n = 292$), scratching the skin to bleeding (34.3 %, $n = 277$), punching to bruising (33.3 %, $n = 269$), hitting the head or other body parts violently to cause bruising (32.4 %, $n = 262$), biting to bleeding (31.7 %, $n = 256$), inserting objects to the nail or skin (29.1 %, $n = 235$), erasing skin (23.1

%, $n = 187$), burning (18.2 %, $n = 147$), scrubbing skin using bleach or cleaner (15.8 %, $n = 128$), and dripping acid onto skin (13.6 %, $n = 110$).

The means, standard deviations, skewness, and kurtosis of each variable, as well as the correlations between them, were shown in Table 1. As expected, attachment anxiety, rumination, and NSSI were significantly and positively correlated. Meaning in life was significantly and negatively correlated with rumination and NSSI, laying the foundation for the establishment of the subsequent buffering effect. Besides, gender was negatively related to attachment anxiety ($b = -0.08$, $p < 0.001$) and rumination ($b = -0.04$, $p < 0.05$), but positively related to meaning in life ($b = 0.06$, $p < 0.001$). Therefore, gender was set as a control variable in the following model testing.

3.2. The mediation model

Next, we tested the mediating role of rumination between attachment anxiety and NSSI. As shown in Fig. 1, attachment anxiety was positively correlated with both NSSI ($b = 0.22$, $SE = 0.02$, $p < 0.001$) and rumination ($b = 0.15$, $SE = 0.02$, $p < 0.001$). Rumination and NSSI were also positively correlated ($b = 0.12$, $SE = 0.02$, $p < 0.001$). The indirect effect from attachment anxiety to NSSI through rumination was significant (indirect effect = 0.02, $SE = 0.00$, 95 % CI = [0.01, 0.03]). These results suggested that rumination played a partial mediating role between attachment anxiety and NSSI.

3.3. The moderated mediation model

Then we tested the proposed moderated mediation model, and the results were shown in Fig. 2. The interaction effect between meaning in life and attachment anxiety ($b = -0.07$, $SE = 0.02$, $p < 0.001$) and that between meaning in life and rumination ($b = -0.06$, $SE = 0.02$, $p < 0.001$) were significant. Meaning in life significantly reduced the impacts of both attachment anxiety and rumination on NSSI. To further interpret the interaction patterns, we conducted the tests of simple slopes. As shown in Fig. 3, the association between attachment anxiety and NSSI was weaker when individuals reported high meaning in life (one standard deviation above the sample mean, $b_{\text{simple}} = 0.11$, $t = 4.57$, $p < 0.001$) than when they reported low meaning in life (one standard deviation below the sample mean, $b_{\text{simple}} = 0.25$, $t = 9.20$, $p < 0.001$). Similarly, the association between rumination and NSSI was weaker when individuals reported high meaning in life (one standard deviation above the sample mean, $b_{\text{simple}} = 0.12$, $t = 5.34$, $p < 0.001$) than when they reported low meaning in life (one standard deviation below the sample mean, $b_{\text{simple}} = 0.25$, $t = 9.27$, $p < 0.001$). The details were shown in Fig. 4.

4. Discussion

In the current study, we proposed a moderated mediation model

Table 1
Descriptive statistics of and correlations between the study variables.

Variables	1	2	3	4	5	6
1. Age	–					
2. Gender	0.07**	–				
3. Attachment Anxiety	–0.00	–0.08**	–			
4. Rumination	–0.00	–0.04*	0.16***	–		
5. Meaning in Life	0.02	0.06**	–0.16**	0.22***	–	
6. NSSI	–0.02	–0.03	0.21**	0.14**	–0.19**	–
<i>M</i>	12.25	–	9.86	24.59	48.70	4.11
<i>SD</i>	0.86	–	5.44	6.21	10.87	9.30
Skewness	0.29	0.36	0.29	0.36	–0.04	3.23
Kurtosis	–0.94	0.10	–0.94	0.10	–0.12	11.52

Note. ** $p < 0.01$, *** $p < 0.001$.

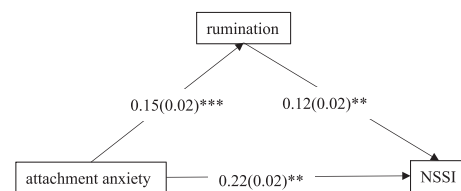


Fig. 1. The mediating effect of rumination in the association between attachment anxiety and NSSI. Note. *** $p < 0.001$; Gender is used as the control variable in this model.

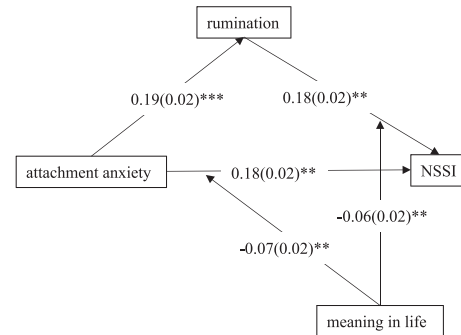


Fig. 2. The moderating effect of meaning in life on associations between attachment anxiety and NSSI. Note. ** $p < 0.01$, *** $p < 0.001$; Gender is used as the control variable in this model.

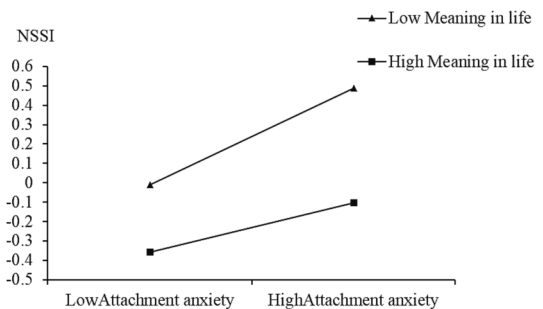


Fig. 3. The moderating effect of meaning in life on the association between attachment anxiety and NSSI.

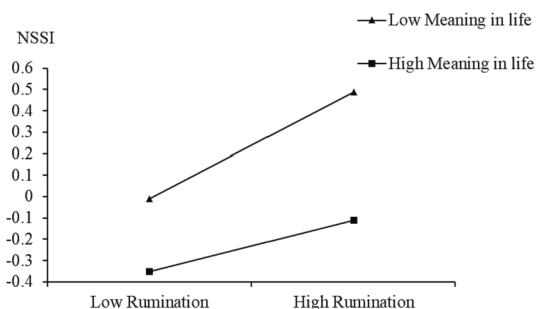


Fig. 4. The moderating effect of meaning in life on the association between rumination and NSSI.

according to the IMV model of NSSI to explain the mechanism underlying the relation between attachment anxiety and NSSI. Consistent with our hypotheses, rumination mediated the relationship between attachment anxiety and NSSI, and meaning in life moderated both the direct relationship and the second stage of the mediating relationship. Next, we will discuss each of the results in detail.

First, individuals with attachment anxiety may be more prone to engage in NSSI. The finding is consistent with previous studies (Woo, Wrath, & Adams, 2020; Michelson & Bhugra, 2012). Individuals with attachment anxiety may adopt NSSI to relieve their anxiety and to seek concerns from their attachment figures, which are respectively supported by the negative intrapersonal reinforcement function and the positive interpersonal reinforcement function of NSSI (Nock, 2010). Besides, NSSI may also be used to punish themselves (Nock, 2010), because they tend to think themselves are not beloved (Bowlby, 1977) and are deserved to be hurt and punished (Hooley et al., 2010). Future research could verify our speculations by exploring the functions of NSSI among individuals with attachment anxiety.

Second, rumination partially mediated the relationship between attachment anxiety and NSSI. This finding is also in line with those of previous research (Garrison et al., 2014; Selby et al., 2013; Selby, Anestis, & Joiner, 2008). Martin and Tesser (1989) has posited that when individuals failed on their important goals, they would engage in rumination to find out the reasons of their failures. For individuals with attachment anxiety, one of their important goals is to establish and maintain intimacy with their attachment figures (Pereg & Mikulincer, 2004). But their need for intimacy is so overwhelming (Brennan et al., 1998) that this goal always fails, leading them to engage in constant rumination (Moreira & Maia, 2021; Shaver & Mikulincer, 2002). Unfortunately, rumination in turn makes them entrapped in negative experiences, and then they would fall into a “rumination - negative emotions” loop (i.e., rumination leads to negative emotions, and negative emotions also lead to increased rumination). To get out of the loop, they may engage in NSSI, because NSSI is “the fastest and most convenient” way that can immediately distract them from the ruminative thinking and psychological pain (Nock, 2010). This finding is also supported by the IMV model, which suggests NSSI could help individuals decrease their aversive thoughts or feelings (O'Connor, 2011; O'Connor et al., 2012).

Another major contribution of the current study is the finding that the direct and indirect relationships between attachment anxiety and NSSI were moderated by meaning in life. Specifically, meaning in life lowered the strength of the association between attachment anxiety and NSSI, and the strength of the association between rumination and NSSI. Although some previous studies showed a negative relationship between meaning in life and NSSI (Marco et al., 2016; Rodríguez et al., 2017), to our knowledge, this is the first empirical article exploring the protective effect of meaning in life on NSSI with a moderated mediation model. This result suggests that although individuals with attachment anxiety would engage in rumination, high meaning in life can help them positively face up the distress and stop them timely from thinking about NSSI (Ying et al., 2020) by providing adaptive resources. Besides, high meaning in life may buffer the risk of NSSI by protecting individuals from the risk factors of NSSI (e.g., depression and anxiety; Schulenberg, 2004; Valencia-Agudo et al., 2018; Taliaferro & Muehlenkamp, 2017). Future studies could further explore how meaning in life affects NSSI.

5. Limitations and future directions

While interpreting the results, the limitations of this study must be considered. First of all, this study is a cross-sectional study. The causal relationships have not been determined. Therefore, longitudinal or quasi-experiment designs could be used in future studies. Second, the present study assessed various variables by self-reports, which may be affected by recall bias and social approval effects. Future researchers can take different assessment methods to supplement the research results. Finally, attachment anxiety in the current study mainly refers to the attachment relationship between adolescents and their parents. Teenagers spend more time with their peers than parents, and friendship is especially important during adolescence (Lu et al., 2015). Peer attachment anxiety may also be an important predictor of NSSI. In future studies, the predictive effects of peer attachment anxiety on NSSI could

be explored.

6. Implications

Despite the aforementioned limitations, the current study investigated a moderated mediation model in Chinese secondary school students and made a meaningful progress in exploring the potential mechanisms underlying the relationship between attachment anxiety and NSSI. Specifically, the examination of the mediating mechanism advances our understanding of the relationship between attachment anxiety and NSSI from a cognitive perspective, and the examination of the moderating mechanism shows that meaning in life weakens the influence of attachment anxiety on NSSI.

In addition to the theoretical implications mentioned above, this study also offers practical guidance. First, parents and clinicians must be concerned about the long-term effects of early attachment anxiety because attachment anxiety may lead to the development of maladaptive cognitive styles (e.g., rumination) and behaviors (e.g., NSSI) in adolescents. Secure parent-child attachment relationships (low attachment anxiety and attachment avoidance) should be encouraged. Second, as a maladaptive cognitive factor that contributes to NSSI, rumination could be spotlighted in the prevention and intervention of NSSI. And the Cognitive Behavior Therapy targeting cognitive vulnerabilities (e.g., rumination) may also be useful in treating NSSI. Third, parents, schools, and even the public should pay attention to the development of adolescents' meaning in life. For example, parents and teachers can expand the horizons of adolescents and provide opportunities for adolescents to find meaning in life. Meanwhile, previous studies have identified mindfulness as an effective way to increase meaning in life (Chu & Mak, 2020; Garland et al., 2015). Thus, teachers and clinicians are encouraged to improve the level of mindfulness in adolescents by some mindfulness programs.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Informed Consent

All the participants provided the informed consent.

Ethical Approval

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

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